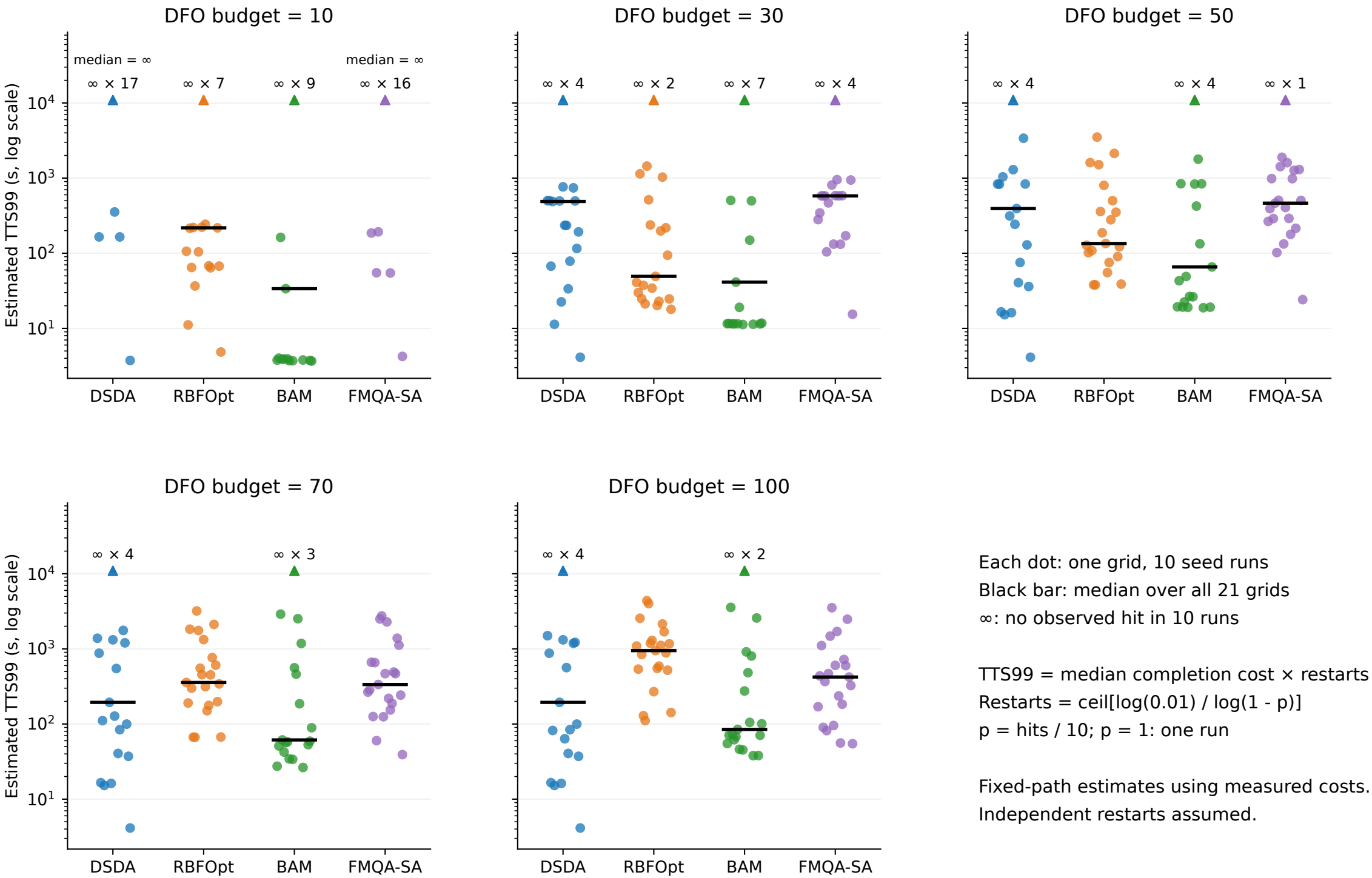
# Grid GDP: estimated TTS99 with measured IPOPT point costs




Each dot: one grid, 10 seed runs
Black bar: median over all 21 grids
∞: no observed hit in 10 runs

TTS99 = median completion cost × restarts
Restarts = ceil[log(0.01) / log(1 - p)]
p = hits / 10; p = 1: one run

Fixed-path estimates using measured costs.
Independent restarts assumed.